

Stage 1 Target preparation and validation

Clean structures, retain catalytic zinc, derive grid box.
Redock reference ligand. Benchmark against matched decoys.



Stage 2 Library construction

Mine ChEMBL for measured binders. Filter on potency,
assay confidence and physicochemical properties.



Stage 3 Primary docking and rescoring

Dock each library against its own target. Rank by affinity
and by ligand efficiency. Apply calibrated threshold.



Stage 4 Cross-docking and dual selection

Dock each surviving set against the opposite target.
Dual actives are the intersection.



Stage 5 Scaffold analysis and enumeration

Extract shared scaffolds. Enforce dual pharmacophore.
Enumerate analogues. Filter on synthetic accessibility.



Stage 6 Antitarget counter-screen

Screen against ACE and DPP-8/9. Penalise cross-reactivity.



Prioritised set for in vitro testing

Selectivity-filtered, with full provenance per compound.